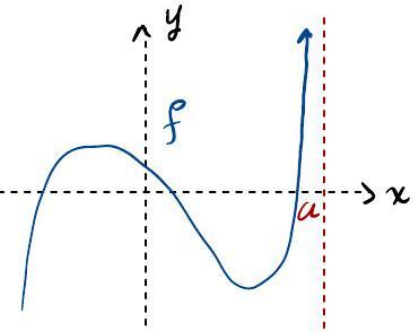
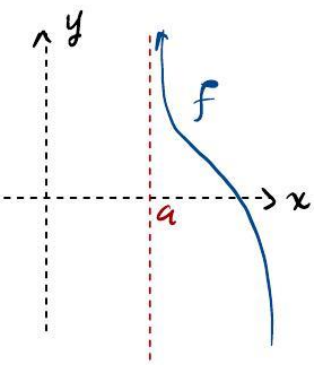
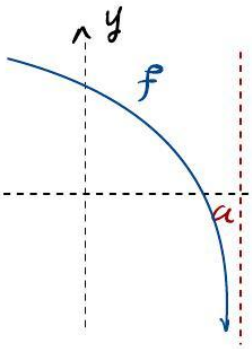
Lecture 3, Dec 19th, 2025



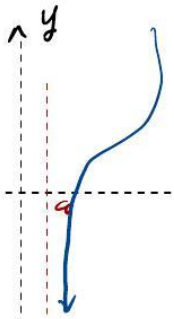
$\lim_{x \to a^-} f(x) = \infty$



$\lim_{x \to a^+} f(x) = \infty$



$\lim_{x \to a^-} f(x) = -\infty$



$\lim_{x \to a^+} f(x) = -\infty$

Def. the vertical line $x = a$ is called a vertical asymtote of the curve $y = f(x)$ if at least one of the following statements is true.

$\lim_{x \to a} f(x) = \infty$ $\quad$ $\lim_{x \to a^-} f(x) = \infty$ $\quad$ $\lim_{x \to a^+} f(x) = \infty$

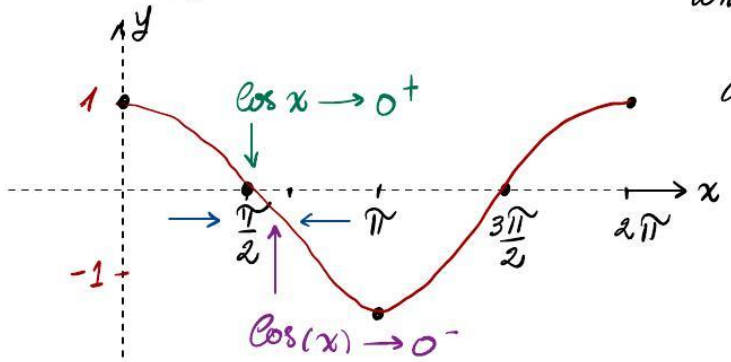
$\lim_{x \to a} f(x) = -\infty$ $\quad$ $\lim_{x \to a^-} f(x) = -\infty$ $\quad$ $\lim_{x \to a^+} f(x) = -\infty$.

Example. Find the vertical asymtotes of $f(x) = \tan x$.

Sln. We know that

$$\tan x = \frac{\sin x}{\cos x}$$

there are potential vertical asymtotes where $\cos x = 0$ for example if $x = \frac{\pi}{2}$.



when $x \to \frac{\pi}{2}^-$ then $\cos x \to 0^+$

and when $x \to \frac{\pi}{2}^+$ then $\cos x \to 0^-$

So, there is a vertical asymtote at $x = \frac{\pi}{2}$.

In general we know that $\cos x = 0$ for $x = \frac{\pi}{2} + n\pi$ where n is an integer, and all are vertical asymtotes of $f(x) = \tan x$.

Example. Find $\lim_{x\to 0} \sin\frac{\pi}{x}$.

Sln. Notice that if we try approximate to 0 we get

$f(1) = \sin \pi = 0$

$f(\frac{1}{2}) = \sin 2\pi = 0$

$f(\frac{1}{3}) = \sin 3\pi = 0$

$f(0.1) = \sin 10\pi = 0$.

$f(0.01) = \sin 100\pi = 0$

We can guess that the limit is 0, but this time our guess is wrong. Notice that if $x = \frac{1}{n}$ then $f(\frac{1}{n}) = \sin n\pi = 0$ for $n \in \mathbb{Z}$.

but if $x = \frac{2}{5}$ then $f(\frac{2}{5}) = \sin(5(\frac{\pi}{2})) = 1$.

$x = \frac{2}{101}$ then $f(\frac{2}{101}) = \sin(101(\frac{\pi}{2})) = 1$

$\vdots$

$x = \frac{2}{\text{odd \#}}$ then $f(\frac{2}{\text{odd \#}}) = 1$

then $\lim_{x\to 0} \sin\frac{\pi}{x}$ does not exist.

Calculating Limits using the Limits Laws.

We use the following properties of limits called the limits laws to calculate limits.

Thm (Limit Laws). Suppose that c is a constant and the limits $\lim_{x\to a} f(x)$ and $\lim_{x\to a} g(x)$ exists. then,

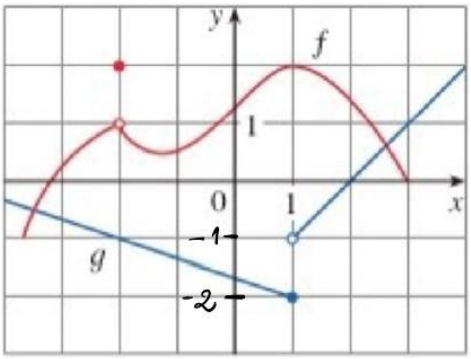
i) $\lim_{x\to a} [f(x) \pm g(x)] = \lim_{x\to a} f(x) \pm \lim_{x\to a} g(x)$

ii) $\lim_{x\to a} [c \cdot f(x)] = c \cdot \lim_{x\to a} f(x)$

iii). $\lim_{x\to a} [f(x)\cdot g(x)] = \left[\lim_{x\to a} f(x)\right]\cdot\left[\lim_{x\to a} g(x)\right]$

iv) $\lim_{x\to a} \frac{f(x)}{g(x)} = \frac{\lim_{x\to a} f(x)}{\lim_{x\to a} g(x)}$, if $\lim_{x\to a} g(x) \neq 0$.

Example.



Compute.

a) $\lim_{x\to -2} [f(x) + 5g(x)]$.

b) $\lim_{x\to 1} [f(x)\cdot g(x)]$.

c) $\lim_{x\to 2} \frac{f(x)}{g(x)}$. For you!

Sln: a) $\lim_{x\to -2} f(x) = 1$ $\lim_{x\to -2} g(x) = -1$.

then we can apply the Law 1 and Law 2.

$$\lim_{x\to -2} [f(x) + 5g(x)] = \lim_{x\to -2} f(x) + \lim_{x\to -2} (5g(x)) \quad \text{by Law 1.}$$

$$= \underbrace{\lim_{x\to -2} f(x)}_{1} + 5\underbrace{\lim_{x\to -2} g(x)}_{-1} \quad \text{by Law 2.}$$

$$= 1 + 5(-1) = -4.$$

b) $\lim_{x\to 1} f(x) = 2$ $\lim_{x\to 1} g(x) = DNE$.

$\lim_{x\to 1^-} f(x) = 2$ $\lim_{x\to 1^-} g(x) = -2$

$\lim_{x\to 1^+} f(x) = 2$ $\lim_{x\to 1^+} g(x) = -1$.

$$\lim_{x\to 1^-} [f(x)\cdot g(x)] = \lim_{x\to 1^-} f(x)\cdot \lim_{x\to 1^-} g(x) = 2(-2) = -4.$$

$$\lim_{x\to 1^+}[f(x)\cdot g(x)] = \lim_{x\to 1^+} f(x)\cdot \lim_{x\to 1^+} g(x) = 2(-1) = -2.$$

Since $\lim_{x\to 1^-}[f(x)\cdot g(x)] \neq \lim_{x\to 1^+}[f(x)\cdot g(x)]$ then

$$\lim_{x\to 1}[f(x)\cdot g(x)] = DNE$$

If we use the product rule repeatedly we obtain the following law.

Power law $\lim_{x\to a}[f(x)]^n = [\lim_{x\to a} f(x)]^n$ where $n\in\mathbb{N}$.

Root Law $\lim_{x\to a}\sqrt[n]{f(x)} = \sqrt[n]{\lim_{x\to a} f(x)}$ where $n\in\mathbb{N}$.

(if n is even, we assume that $\lim_{x\to a} f(x) \geq 0$.)

- $\lim_{x\to a} c = c$
- $\lim_{x\to a} x = a$
- $\lim_{x\to a} x^n = a^n$, $a\in\mathbb{N}$.
- $\lim_{x\to a}\sqrt[n]{x} = \sqrt[n]{a}$, $n\in\mathbb{N}$ if n is even, $a\geq 0$.

Example. Evaluate the limits and justify each step.

a). $\lim_{x\to 5}(2x^2-3x+4)$ b). $\lim_{x\to 2}\dfrac{x^3+2x^2-1}{5-3x}$.

Sln. a). $\lim_{x\to 5}(2x^2-3x+4) = \lim_{x\to 5}(2x^2) - \lim_{x\to 5}(3x) + \lim_{x\to 5} 4$ by law 1.

$= 2\lim_{x\to 5} x^2 - 3\lim_{x\to 5} x + \lim_{x\to 5} 4$ by law 2.

$\lim_{x\to a} x^2 = a^2$ $\lim_{x\to a} x = a$ $\lim_{x\to a} c = c$.

$= 2(5)^2 - 3(5) + 4 = 39$.

b). • $\lim_{x\to 2}(5-3x) = \lim_{x\to 2} 5 - 3\lim_{x\to 2} x = 5 - 3(2) = -1 \neq 0$.

Using the Law 4.

$$\lim_{x\to 2} \frac{x^3+2x^2-1}{5-3x} = \frac{\lim_{x\to 2}(x^3+2x^2-1)}{\lim_{x\to 2}(5-3x)} = \frac{15}{-1} = -15.$$

Sometimes is easy compute the limits by Direct substitution.

Notice that

$$\lim_{x\to 2} x^3+2x^2-1 = 15 \quad \text{but if} \quad f(x) = x^3+2x^2-1$$

$$f(2) = 2^3+2(2)^2-1 = 15.$$

that is $\lim_{x\to 2} f(x) = f(2)$.

Direct Substitution Property. If f is a polynomial or a rational function and a is in the Domain of f, then

$$\lim_{x\to a} f(x) = f(a).$$

Rmk. Functions that have the Direct Substitution property are called continuous at a.

Example. Find $\lim_{x\to 1} \frac{x^2-1}{x-1}$.

Sln. Let $f(x) = \frac{x^2-1}{x-1}$ $\quad Dom f = (-\infty,1)\cup(1,\infty)$, we can't find the limit by substituting $x=1$ b/c $f(1)$ is not defined. Now, we can't apply the Quotien Law b/c the limit of the denominator is 0. Instead, we need to do some preliminary algebra.

if $x \neq 1$ then $\frac{x^2-1}{(x-1)} = \frac{(x-1)(x+1)}{(x-1)}$ since $x \neq 1$

$$\lim_{x\to 1} \frac{x^2-1}{x-1} = \lim_{x\to 1}(x+1) = 2.$$

Example. Evaluate $\lim_{h\to 0} \frac{(3+h)^2-9}{h}$.

Sln. Let $F(h) = \dfrac{(3+h)^2 - 9}{h}$ we know $(x+y)^2 = x^2 + 2xy + y^2$

$$= \frac{9 + 6h + h^2 - 9}{h} = \frac{6h + h^2}{h} = \frac{h(6+h)}{h} = 6 + h.$$

that means

$$\lim_{h \to 0} F(h) = \lim_{h \to 0} 6 + h = 6.$$

Example. Find $\lim_{t \to 0} \dfrac{\sqrt{t^2+9} - 3}{t^2}$.

Sln. We want apply $(a^2 - b^2) = (a-b)(a+b)$.

if you have $a+b \Rightarrow a-b$ is the conjugate

if you have $a-b \Rightarrow a+b$ is the conjugate

"the idea is rationalize the numerator"

$\underbrace{\sqrt{t^2+9}}_{a} \quad \underbrace{-3}_{b}$ then $\sqrt{t^2+9} + 3$ is the conjugate.

then $(\sqrt{t^2+9} - 3)(\sqrt{t^2+9} + 3) = (\sqrt{t^2+9})^2 - (3)^2 = t^2 + 9 - 9 = t^2$ (IT)

$$\lim_{t \to 0} \frac{\sqrt{t^2+9} - 3}{t^2} = \lim_{t \to 0} \frac{\sqrt{t^2+9} - 3}{t^2} \times \boxed{\frac{\sqrt{t^2+9} + 3}{\sqrt{t^2+9} + 3}}$$

$$= \lim_{t \to 0} \frac{(\sqrt{t^2+9} - 3)(\sqrt{t^2+9} + 3)}{t^2(\sqrt{t^2+9} + 3)} \qquad (\text{numerator} = 1)$$

$$= \lim_{t \to 0} \frac{t^2}{t^2(\sqrt{t^2+9} + 3)} \quad \text{by (IT)}.$$

$$= \lim_{t \to 0} \frac{1}{\sqrt{t^2+9} + 3} = \frac{1}{\sqrt{0^2+9} + 3} = \frac{1}{3+3} = \frac{1}{6}.$$

Using One-sided Limits

Example. Show that $\lim_{x\to 0} |x| = 0$

Sln. We know that

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

So, $\lim_{x\to 0^+} |x| = \lim_{x\to 0^+} x = 0.$

$\lim_{x\to 0^-} |x| = \lim_{x\to 0^-} (-x) = -\lim_{x\to 0} x = 0.$

then by thm ♡ $\lim_{x\to 0} |x| = 0.$

Example. If

$$f(x) = \begin{cases} \sqrt{x-4} & \text{if } x > 4 \\ 8-2x & \text{if } x < 4. \end{cases}$$

determine whether $\lim_{x\to 4} f(x)$ exists.

Sln. $\lim_{x\to 4^-} f(x) = \lim_{x\to 4^-} (8-2x) = 8-2(4) = 0.$

$\lim_{x\to 4^+} f(x) = \lim_{x\to 4^+} \sqrt{x-4} = 0$

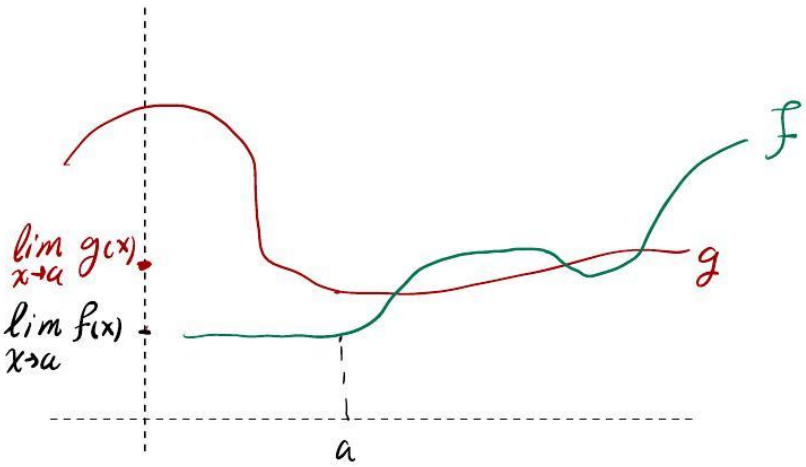
by thm ♡ since right-left hand limits are equal then

$$\lim_{x\to 4} f(x) = 0.$$

Problem. Compute $\lim_{x\to 0} x^2 \cos\left(\frac{1}{x}\right)$

The squeeze theorem The following theorems describe how the limit of functions are related when the values of one function are greater than those of another.

Thm 1 If $f(x) \leq g(x)$ when x is near a and the limits of f and g exist as x approaches a, then.



$$\lim_{x \to a} f(x) \leq \lim_{x \to a} g(x).$$